

BARYON ASYMMETRY AS CURVATURE-RATIO LOCK-IN: RATIO-LEVEL FIRST PRINCIPLES IN HQIV-LEAN

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ABSTRACT. We state the strongest *ratio-level* first-principles statement for baryon asymmetry currently supported by the HQIV-LEAN library, positioned after the trilogy on rapidity/Spin(8) closure [1], three-dimensional null-shell growth and octonionic gauge packaging [2], and the discrete octonionic action layer [3]. The formal spine is entirely **ratio-level**: discrete shell combinatorics define a cumulative curvature integral $I(n)$; horizon ratios $\Omega_k(n; N) = I(n)/I(N)$ are machine-checked with proved monotonicity in n at fixed N ; a distinguished lock-in shell m_{lockin} (the library pin **referenceM**) calibrates $\Omega_k = 1$ at the lock-in horizon; and the baryon-asymmetry readout $\eta(n; N)$ is **defined** to share that same normalization, pointwise in (n, N) , with an optional single anchor for the absolute scale. Big-bang nucleosynthesis is **out of scope** as a Lean theorem: standard BBN takes the same η convention fixed at lock-in as input and computes light-element abundances via external Einstein–Boltzmann and nucleosynthesis networks; verifying those abundances against observation is numerical physics in companion pipelines, not part of the proof graph traced here. What we *do* claim, with file-level precision, is that baryogenesis bookkeeping is not an independent knob: it is the same partial curvature imprint that appears in the growth channel, triality-sensitive CP-bias scaffolds, and the HQVM Friedmann compatibility package—with explicit honesty boundaries where SM transport, washout, and sphaleron dynamics remain narrative targets rather than formalized theorems.

1. INTRODUCTION AND PLACEMENT

The HQIV programme ties discrete null-shell counting, a curvature imprint exponent $\alpha = 3/5$ (lattice-coherent with Brodie’s horizon thermodynamics [5]), and an eight-dimensional octonionic carrier with machine-checked $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$ data [1, 2]. The companion action manuscript [3] documents how the same (α, ϕ) slots feed discrete O–Maxwell stationarity and an HQVM Friedmann *compatibility* identity.

This note answers a narrower question: **what baryogenesis statement is already a theorem in HQIV-LEAN, and what is still interpretive physics?** We use the library’s own quarantine discipline: geometry and ratios live in `Hqiv/Physics/BaryogenesisCore` and `Hqiv/Geometry/OctonionicLightCone`; the observed-order η anchor is isolated in `Hqiv/Physics/BaryogenesisEtaPaper` and wired only through `Hqiv/Physics/BaryogenesisWitness`.

2. READER CONTRACT (FOUR TIERS)

(I) Proved in Lean (this paper’s formal spine): Shell indices m_{QCD} , m_{lockin} , ladder temperatures T_{QCD} , T_{lockin} , combinatorial curvature norm $\mathcal{N}_{\text{curv}}$, per-shell imprint $\delta_E(m)$, cumulative integral $I(n) = \sum_{m < n} |\sigma(m)|$ (implemented as `curvature_integral`), strict monotonicity $I(n_1) < I(n_2)$ for $n_1 < n_2$ (`curvature_integral_strict_mono`), horizon ratios $\Omega_k(n; N)$, monotonicity $\Omega_k(n_1; N) \leq \Omega_k(n_2; N)$ for $n_1 \leq n_2$ (`omega_k_at_horizon_mono`) with strict version (`omega_k_at_horizon_strict_mono`), bound $\Omega_k(n; N) \leq 1$ when $n \leq N$ (`omega_k_at_horizon_le_one_of_le`), lock-in calibration $\Omega_k(m_{\text{lockin}}; m_{\text{lockin}}) = 1$, parallel $\eta(n; N)$ with the same monotonicity and strict monotonicity (`eta_at_horizon_mono`, `eta_at_horizon_strict_mono`), bound $\eta(n; N) \leq \eta_{\text{paper}}$ when $n \leq N$ (`eta_at_horizon`

`_le_eta_paper`), and the identity $\eta = \eta_{\text{paper}} \cdot \Omega_k$ at fixed horizon when denominators are positive.

- (II) **Proved algebraic packaging (still Lean).**: Triality/representation-sensitive increments built from the curvature-ratio bias `rapidityCPBias` in `Hqiv/Physics/TrialityRapidityWellEquivalence`; explicit averaging of the affine tilt factors $(1 + w_{\text{rep}} \cdot \text{rapidityCPBias})$ (`triality_cp_tilt_factors_average_eq_one`); and three-representation averaging of the full increments back to the rapidity well (`cpSensitiveTrialityIncrement_threeRep_average_eq_rapidityWell`).
- (III) **Structural compatibility (Lean, not transport).**: `Hqiv/Physics/GRFromMaxwell`: shared ϕ , shared α , and the homogeneous HQVM Friedmann equation packaged as an explicit algebraic equivalence under stated hypotheses—a compatibility bridge to cosmology, not a derivation of washout-limited baryon yield.
- (IV) **Continuum dynamics and SM numerics (not Lean).**: Associator/vorticity sources, QCD window maximisation, sphaleron rates, and nucleosynthesis networks remain companion-manuscript and code-layer material [4].

3. DISCRETE SHELLS, TEMPERATURES, AND THE BARYOGENESIS WINDOW

Fix the HQIV discrete shell index $m \in \mathbb{N}$ as in [1, 2]. The library defines a QCD onset shell index m_{QCD} and a lock-in shell

$$m_{\text{lockin}} := m_{\text{QCD}} + \text{stepsFromQCDToLockin},$$

currently aligned with `referenceM` (the same proton-anchor convention used throughout the formalisation). Temperatures on the ladder are

$$T(m) = \frac{1}{m+1}$$

in the library’s natural-unit convention (`T_QCD_closed`, `T_lockin_closed` in `BaryogenesisCore`).

The finite shell set used for discrete bookkeeping is `baryogenesisShells` (an explicit `Finset` interval from m_{QCD} through a few shells after lock-in). This is the honest discrete substitute for a continuum “GeV window”: it is *not* proved from washout maximisation in Lean; it is the packaged combinatorial stage on which curvature ratios are evaluated.

4. CURVATURE IMPRINT, $\mathcal{N}_{\text{curv}}$, AND δ_E

Let $\sigma(m)$ denote the shell-shape factor (the library’s `shell_shape`), built from the same α -weighted logarithmic curvature density used in the growth trilogy. The combinatorial curvature norm is the exact real scalar

$$\mathcal{N}_{\text{curv}} = 6^7 \sqrt{3},$$

proved in `OctonionicLightCone` from the structural “six cube directions, seven imaginary octonion directions, half-diagonal $\sqrt{3}$ ” bookkeeping (see `curvature_norm_determined_by_structure` and related lemmas). The per-shell imprint is

$$\delta_E(m) = \mathcal{N}_{\text{curv}} \cdot \sigma(m),$$

(`deltaE_at_QCD_shell` specialises the same pattern at m_{QCD}).

Structural rigidity (octonionic vs. quaternionic comparison). Lean also proves that a natural quaternionic Maxwell comparison norm undershoots the HQIV octonionic norm by a rigid factor 1296 shellwise (`deltaE_eq_1296_mul_deltaE_quaternionicCandidate` and neighbours). This supports the narrative that $\mathcal{N}_{\text{curv}}$ is not an arbitrary normalisation: it is the imprint constant selected by the octonionic lift inside the same eight-dimensional carrier used for gauge closure.

5. HORIZON RATIOS Ω_k AND LOCK-IN CALIBRATION

Define the cumulative partial integral

$$I(n) := \sum_{m=0}^{n-1} |\sigma(m)|$$

(`curvature_integral`). For horizons N with $I(N) > 0$, the library defines

$$\Omega_k(n; N) := \frac{I(n)}{I(N)}.$$

At the lock-in shell, Lean proves positivity $I(m_{\text{lockin}}) > 0$ and the calibration

$$\Omega_k(m_{\text{lockin}}; m_{\text{lockin}}) = 1$$

(`omega_k_lockin_calibration`). This is the precise sense in which “spatial curvature lock-in” is a **theorem**: it is normalization of the discrete imprint integral at the declared reference horizon, not an experimental input.

Proposition 5.1 (Monotonicity and approach to lock-in, Lean). *Assume $I(N) > 0$. Then for all $n_1 \leq n_2$, $\Omega_k(n_1; N) \leq \Omega_k(n_2; N)$, with strict inequality when $n_1 < n_2$ (`omega_k_at_horizon_mono`, `omega_k_at_horizon_strict_mono`). If additionally $n \leq N$, then $\Omega_k(n; N) \leq 1$ (`omega_k_at_horizon_le_one_of_le`). In particular, along shells $n \leq m_{\text{lockin}}$ with horizon $N = m_{\text{lockin}}$, the ratio $\Omega_k(n; m_{\text{lockin}})$ increases strictly toward the calibration value 1 as n approaches m_{lockin} from below.*

6. BARYON ASYMMETRY η AS THE SAME NORMALIZATION (OPTIONAL ABSOLUTE ANCHOR)

Without importing any experimental η . Everything in `BaryogenesisCore` is logically independent of the PDG-scale baryon asymmetry.

With a single absolute anchor (quarantined). The constant η_{paper} appears *only* in `BaryogenesisEtaPaper`. The witness module defines

$$\eta(n; N) := \eta_{\text{paper}} \cdot \frac{I(n)}{I(N)}$$

when $I(N) > 0$ (and a conservative fallback otherwise), and proves the parallel identities to the Ω_k ratio (`eta_at_horizon_eq`, `eta_at_horizon_self` at $n = N$).

The conceptual punchline is already a theorem:

$$\eta(n; N) = \eta_{\text{paper}} \cdot \Omega_k(n; N)$$

(`baryogenesis_same_normalization_as_omega_k`). So, **at fixed horizon**, baryogenesis is not a second independent imprint law in Lean: it is the same partial-integral ratio as Ω_k , optionally scaled by one overall anchor.

Same monotonicity and bounds as Ω_k . Lean also proves $\eta(n_1; N) \leq \eta(n_2; N)$ for $n_1 \leq n_2$ (`eta_at_horizon_mono`), strict increase when $n_1 < n_2$ and $I(N) > 0$ (`eta_at_horizon_strict_mono`), and $\eta(n; N) \leq \eta_{\text{paper}}$ whenever $n \leq N$ (`eta_at_horizon_le_eta_paper`). Thus the “optional single anchor” is both an overall scale and an *upper envelope* for readouts inside the horizon. Thus η_{paper} functions simultaneously as the absolute scale (when read at the lock-in horizon) and as a strict upper bound on the asymmetry that can be read out at any earlier shell—a direct consequence of the shared monotonicity with Ω_k .

Honesty boundary (“first principles” vs. calibration). Lean does not currently derive η_{paper} from associator phases or SM chemical potentials. What *is* first-principles inside the formalisation is the **shared ratio law** and the **lock-in calibration** of Ω_k . Attaching the observed η scale is therefore explicitly labeled as a *single-point calibration* imported into `BaryogenesisEtaPaper`, analogous in spirit to how the companion HQIV physics manuscript treats overall amplitude normalisation while keeping imprint *shape* combinatorially rigid.

7. CP BIAS WITHOUT NAMING η : TRIALITY COUPLING

The module `TrialityRapidityWellEquivalence` defines

$$\text{rapidityCPBias}(m) := \Omega_k(m; m_{\text{lockin}}) - 1,$$

a pure curvature-ratio deviation at the lock-in reference horizon. It then builds representation-sensitive triality increments by multiplying baseline barrier surfaces by $(1 + \text{rapidityCPBias}(m) \cdot w_{\text{rep}})$ with explicit $w_{\text{rep}} \in \{0, +1, -1\}$ on the three $\text{Spin}(8)$ triality channels, and proves exact three-rep averaging back to the rapidity well.

Auditable zero-sum tilt. Writing the weights as $w_{8v} = 0$, $w_{8s+} = +1$, $w_{8s-} = -1$, Lean proves the affine factors average identically to unity:

$$(1) \quad \frac{1}{3} \sum_{\text{rep} \in \{8v, 8s+, 8s-\}} (1 + w_{\text{rep}} \cdot \text{rapidityCPBias}(m)) = 1$$

(`triality_cp_tilt_factors_average_eq_one`). The full triality increments `cpSensitiveTrialityIncrement` average to the rapidity-written well surface `rapidityLiftedWell` (`cpSensitiveTrialityIncrement_threeRep_average_eq_rapidityWell`).

This is the cleanest Lean-backed “CP-odd bias is tied to the same curvature-ratio channel” statement that does *not* import η_{paper} . It is still **not** a replacement for CPT/sphaleron/washout dynamics; it is a structured, finite-dimensional coupling scaffold consistent with the lock-in ratios.

8. LINK TO HQVM FRIEDMANN AND O-MAXWELL (COMPATIBILITY, NOT YIELD)

`GRFromMaxwell` packages that the same auxiliary ϕ appears in the HQVM lapse story and the O-Maxwell bookkeeping (`same_phi_in_O_Maxwell_and_HQVM`), and that the same lattice $\alpha = 3/5$ is the exponent entering $G_{\text{eff}}(\phi) = \phi^\alpha$ on the Friedmann side (`same_alpha_in_O_Maxwell_and_HQVM`). The homogeneous Friedmann identity is packaged as an explicit algebraic equivalence (`O_Maxwell_compatible_with_HQVM_GR_homogeneous`).

This supports the *narrative* that cosmological background evolution used in companion numerics is not stitched from unrelated parameters: it is the same (α, ϕ) ledger as the discrete gauge sector [3]. It does *not*, by itself, integrate Boltzmann equations for baryon number violation in the SM.

9. WHAT WOULD COMPLETE A “FULL DYNAMICAL DERIVATION”

To go beyond ratio bookkeeping and triality scaffolds, a future formalisation would need explicit discrete or continuum transport theorems, for example:

- a well-posed chemical-potential / Boltzmann layer tied to the same shell indices;
- a washout bound stated as a theorem about monotone functionals of $I(n)$ under a declared interaction kernel;
- or a provable map from non-associative phase holonomy to a signed baryon production density, with finite totals controlled by $I(n)$.

None of these are discharged in the current repository; the present manuscript is therefore the correct “maximal honest” baryogenesis paper aligned to actual Lean artifacts.

BIG-BANG NUCLEOSYNTHESIS (ONE PARAGRAPH, OUT OF SCOPE)

Standard BBN takes the same η convention fixed at lock-in as input and computes light-element abundances via external Einstein–Boltzmann and nucleosynthesis networks. Verifying those abundances against observation is numerical physics performed in companion pipelines; it is not part of the Lean proof graph traced in this note.

10. CONCLUSION

The first-principles content formalised today is:

- (1) a rigid combinatorial imprint norm $\mathcal{N}_{\text{curv}} = 6^7\sqrt{3}$ with an octonionic/quaternionic contrast theorem;
- (2) a discrete cumulative curvature integral $I(n)$ and horizon ratios $\Omega_k(n; N)$ with proved strict monotonicity of I , monotonicity (and strict monotonicity) of Ω_k in the readout shell, the bound $\Omega_k(n; N) \leq 1$ for $n \leq N$, and lock-in calibration $\Omega_k = 1$ at m_{lockin} ;
- (3) a parallel $\eta(n; N)$ law sharing the *identical* normalization as $\Omega_k(n; N)$, with matching monotonicity lemmas and $\eta(n; N) \leq \eta_{\text{paper}}$ for $n \leq N$, and with η_{paper} quarantined to a single module;
- (4) a triality representation scaffold coupling curvature-ratio bias to CP-sensitive increments, without importing η_{paper} . This scaffold ties CP-odd increments to the same curvature-ratio channel used for Spin(8) closure and measurement; it does *not* derive the bias itself from non-associativity of the octonion product (that remains a Tier-IV target).
- (5) a GR/O–Maxwell compatibility package tying (α, ϕ) across sectors.

That package is already enough for a mathematically disciplined “baryogenesis stands on the same shoulders as Spin(8) closure” paper; the remaining distance to textbook electroweak baryogenesis is the gap between ratio-level bookkeeping (already formalised) and yield-level dynamics (washout, sphaleron integration, and associator-derived CP violation), all of which remain outside the current Lean corpus.

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APPENDIX A. LEAN DEPENDENCY SKETCH

OctonionicLightCone

```
(shell_shape, curvature_integral, curvature_integral_strict_mono,
 omega_k_at_horizon_mono / _strict_mono / _le_one_of_le,
 curvature_norm_combinatorial, deltaE, referenceM, qcdShell, ...)
```

|
v

BaryogenesisCore

```

(m_QCD, m_lockin, T_QCD, T_lockin, baryogenesisShells,
 omega_k_lockin_calibration, baryogenesis_vital_omega_T_no_eta)
|
+-----> TrialityRapidityWellEquivalence
|
|          (rapidityCPBias from omega_k_at_horizon _ m_lockin;
|          triality_cp_tilt_factors_average_eq_one;
|          cpSensitiveTrialityIncrement_threeRep_average_eq_rapidityWel
|
+--> BaryogenesisEtaPaper (eta_paper only)
|
v

```

BaryogenesisWitness

```

(eta_at_horizon_mono, eta_at_horizon_strict_mono, eta_at_horizon_le_eta_paper,
 baryogenesis_same_normalization_as_omega_k, ...)

```

GRFromMaxwell (same_phi / same_alpha / HQVM Friedmann compatibility; parallel 0--Maxwell story

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